

Milestone 1: Data Integration & Data Modeling

Project Introduction

Urban Pulse is a Smart City Mobility Intelligence Platform that unifies transit, ride-hailing, micromobility, weather, and demographic data to help city planners understand and improve urban movement.

Milestone 1 Objective · Weeks 1-2

- Integrate raw mobility, weather & demographic data sources
- Clean, standardize, and validate incoming records
- Design a star-schema data model for analytics
- Prepare a trusted dataset ready for Milestone 2 (Power BI dashboards)



Duration: Weeks 1 – 2

Urban Pulse — Architecture & Data Flow



Data Sources

Transit · Ride-hail · Bike-share · Weather · Demographics



ETL & Cleaning

Ingest · Standardize · Validate



Star Schema Model

Facts + Dimensions



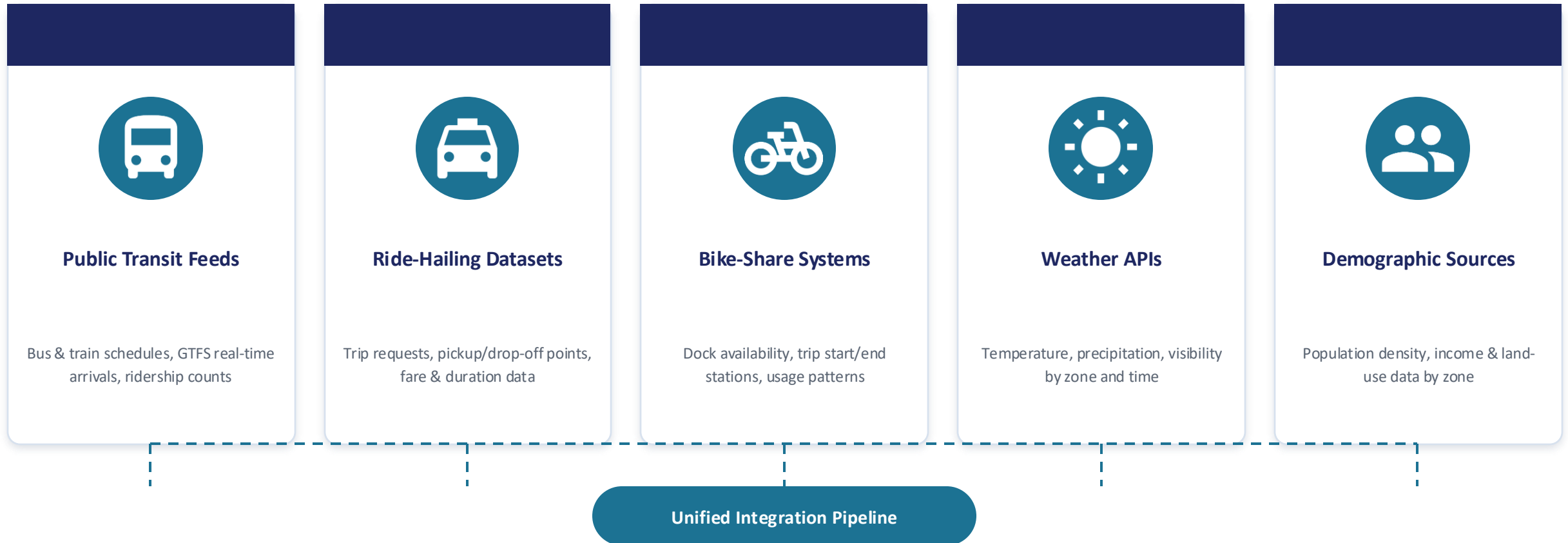
Power BI Analytics

Dashboards & KPIs (Milestone 2)

Data Sources

MILESTONE 1

Urban Pulse ingests five categories of real-world mobility and contextual data, each connected into a unified pipeline.



ETL & Data Cleaning

MILESTONE 1



Key ETL & Cleaning Steps



Data Ingestion

Collect raw files & API streams into a staging layer



Data Standardization

Align field names, units, and formats across sources



Timestamp Standardization

Convert all timestamps to a single time zone/format



Location/Zone Standardization

Map coordinates to consistent city zone IDs



Missing-Value Handling

Impute or flag incomplete records so gaps don't skew analysis



Duplicate Removal

Detect and remove repeated trip/event entries



Data Validation

Run rule-based checks before loading to the model

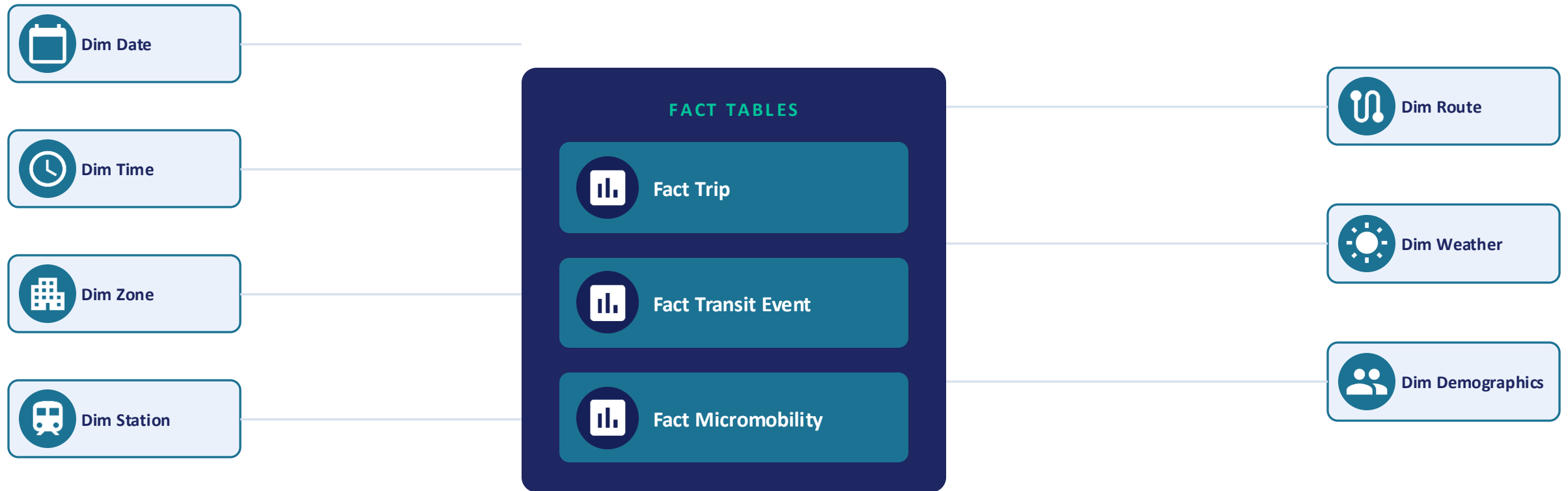


~93% reduction

in duplicate and error records after cleaning

Star Schema Data Model

MILESTONE 1



Fact Tables

Store measurable events — trips, transit events, micromobility rides — with keys linking to dimensions.



Dimension Tables

Provide descriptive context — date, time, zone, station, route, weather, demographics — for filtering and analysis.

Power BI Data Quality & Milestone Result

**2.4M**

Total Records

**5**

Data Sources Integrated

**42**

Zones Covered

**98.6%**

Data Quality

**-93%**

Duplicate / Error Reduction

Milestone 1 Outcome

- ✓ All 5 mobility & contextual data sources successfully integrated
- ✓ Raw data cleaned, standardized, and de-duplicated
- ✓ Validated dataset organized into a star-schema model
- ✓ Dataset is production-ready for Power BI dashboard development in Milestone 2

Data Quality Improvement

